Envoy FPGA crypto: implementation and validation

2026-10-18 Run batch: `envoy-fpga-20261018-rtl-live`

The requested system offloads Envoy's classical and post-quantum operations to AWS F2. This iteration adds actual hash/arithmetic RTL and checks the linked Envoy boundary. Full FPGA signature/KEM controllers, mailbox and F2 shell integration, and physical deployment remain unfinished. No CPU/FPGA speedup or algorithm runtime is measured.

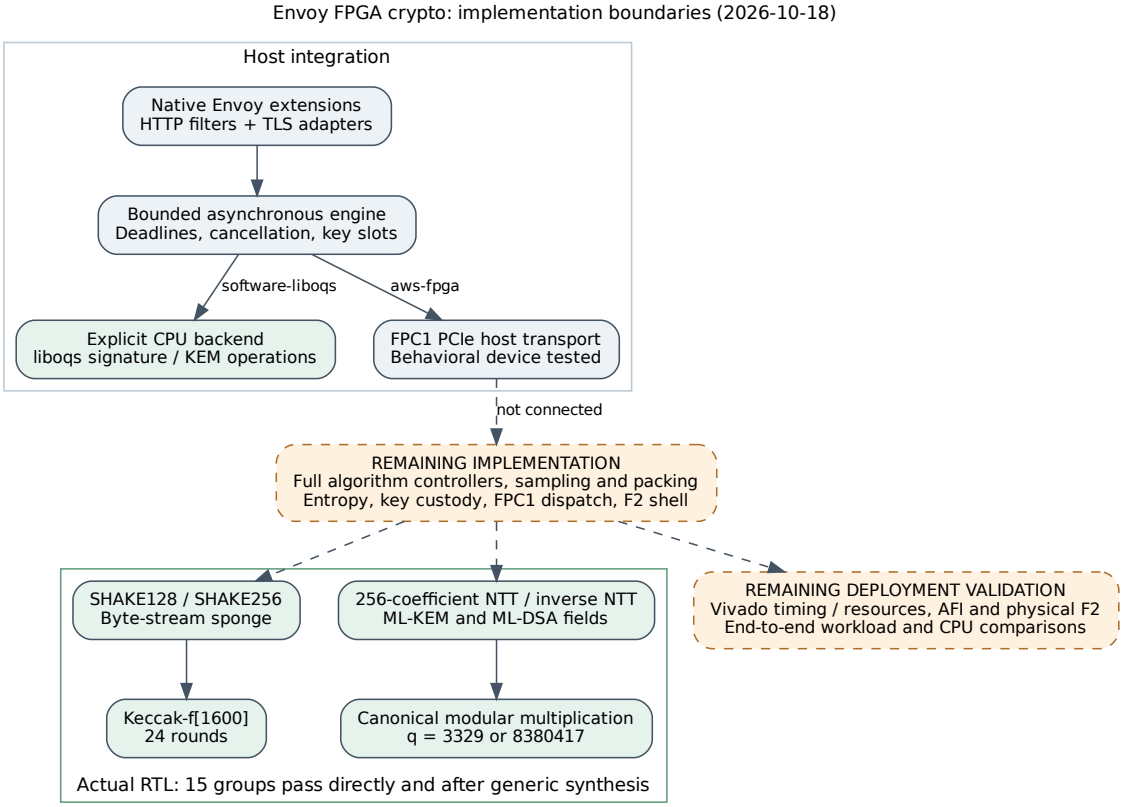


Figure 1. Solid connections are implemented; dashed connections require implementation or deployment validation. Functional test groups are not performance samples, so this explanatory diagram replaces a timing comparison. Source: `architecture.dot` in this report's directory.

Source identity. Extends commit `3c81607c87c4b896e78a49eb6b7f252c0131a96e`, PR #2122. Final project hashes and aggregate outcomes are in the Oct 18 receipt manifest. The owner-approved raw archive and detailed run manifest preserve the original validation batch. The detailed manifest identifies the execution-era snapshot; the summary includes later documentation/PDF updates.

Evidence and reproduction

Validation commands and exact test scope; RTL interface contracts and FIPS 202/203/204 references; Oct 18 receipt archive; full source-linked report. The report and diagram are generated with Graphviz and pdfLaTeX; their editable sources are committed together.

Observed correctness results

Check	Observed result and scope
Source RTL	Fatal Verilator lint passed; 15/15 native C++ groups passed.
Generic synthesis	Structural checks passed; zero combinational loops; no runtime divider, modulo or latch cells.
Generated Verilog	Same 15/15 external groups passed; not a formal equivalence proof or 30 independent test specifications.
Host regressions	36/36 core, classical TLS, PQ and PQ TLS groups passed again.
Linked Envoy	Minimal Envoy v1.39.3 executable linked with the extension.
Live HTTP	5/5 groups passed using CPU liboqs: ML-DSA-65, ML-KEM-768, independent BoringSSL checks, contexts/256 KiB input, rejection and erasure.
Workflow lint	Local actionlint 1.7.7 passed; optional linked Envoy and BoringSSL/liboqs tests are excluded from this scoped CI job.
Root Rust gate	Failed before tests: 26 compilation errors in unchanged base sources, including duplicate `Fp4` variants.

Source simulation enables assertions and randomized register initialization. Public fixture seed: `0x4650433252544c`. SHAKE is checked against OpenSSL; NTT is checked against direct evaluation/interpolation and schoolbook negacyclic convolution. The test's pointwise/base multiplication is C++ oracle code, not a hardware command. Cancellation, invalid operands, backpressure, erasure and concurrent independent engines are exercised.

The generic hierarchy has 14,953 coarse cells, not Xilinx utilization. Writable arrays became registers. This establishes no frequency, physical DSP/BRAM mapping, power, masking, fault resistance or side-channel result.

Failures, environment and remaining scope

The original object-only Envoy check missed a runtime registration collision: both HTTP factories returned generic `google.protobuf.Struct`. Distinct `FpgaHttp` and `PqHttp` configurations now give the factories unique types. Compilation alone did not establish startup. Successful linked-process startup now exercises both registrations.

The first HTTP run passed 2/5 groups because the harness incorrectly required content type on empty acknowledgments. Envoy removes that header. Correcting the harness gave 5/5 without changing the production filter. Nonempty content types, status, body, cache control and backend labels remain checked.

Intermediate attempts failed on width/unused-signal lint, vendor C++ warning settings, generated netlist warnings, one script edited during execution, and an oversized generated model killed during compilation. Logs are retained. The final flow copies its script and uses byte-addressed SHAKE storage; both RTL representations pass. Failed attempts used intermediate sources rather than the final source hash set. Intermediate snapshots and exact initial standalone lint commands were not frozen; final sources and commands are recorded.

Linux x86-64 shared workspace: four CPU quota cores, 16 GiB memory ceiling; GCC 14.2.0, Verilator 5.032, sv2v 0.0.13, Yosys 0.52. There is no isolated benchmark, Vivado, attached F2 or AWS identity. Build diagnostics are not cryptographic performance measurements.

Complete ML-KEM/ML-DSA controllers, sampling, packing, hardware polynomial products, entropy, key custody and FPC1 dispatch remain. Classical hardware cores, an AFI, final FN-DSA and comprehensive FPGA scheme coverage are absent. Stateful LMS/XMSS are also absent. The 262-mode liboqs catalog is an explicit CPU backend. Oct 17 exhaustive catalog and sanitizer results are historical and were not rerun in this batch. TLS KEM key shares remain BoringSSL CPU work; the private-key signing callback does not offload them. Actual Envoy TLS/F2 listeners and the default full extension selection were not tested; live HTTP used ephemeral loopback ports and an explicit CPU backend.

The index-calculus scoreboards, timeline, curve diagrams and performance charts remain unchanged: this change yields no corresponding curve/ECDLP finding or performance measurement.